The Pharynx

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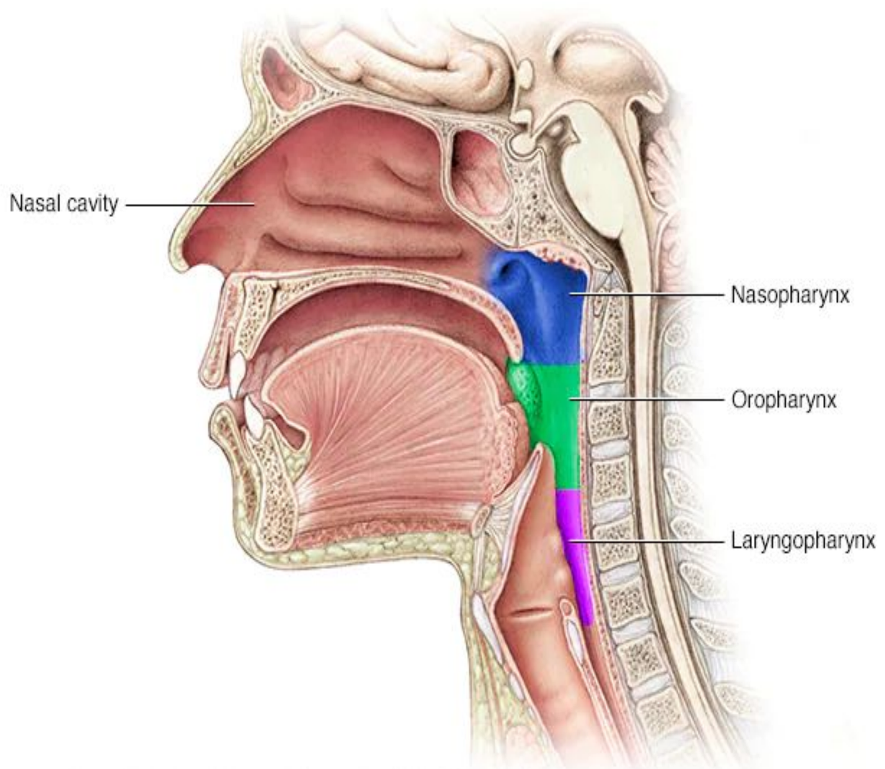
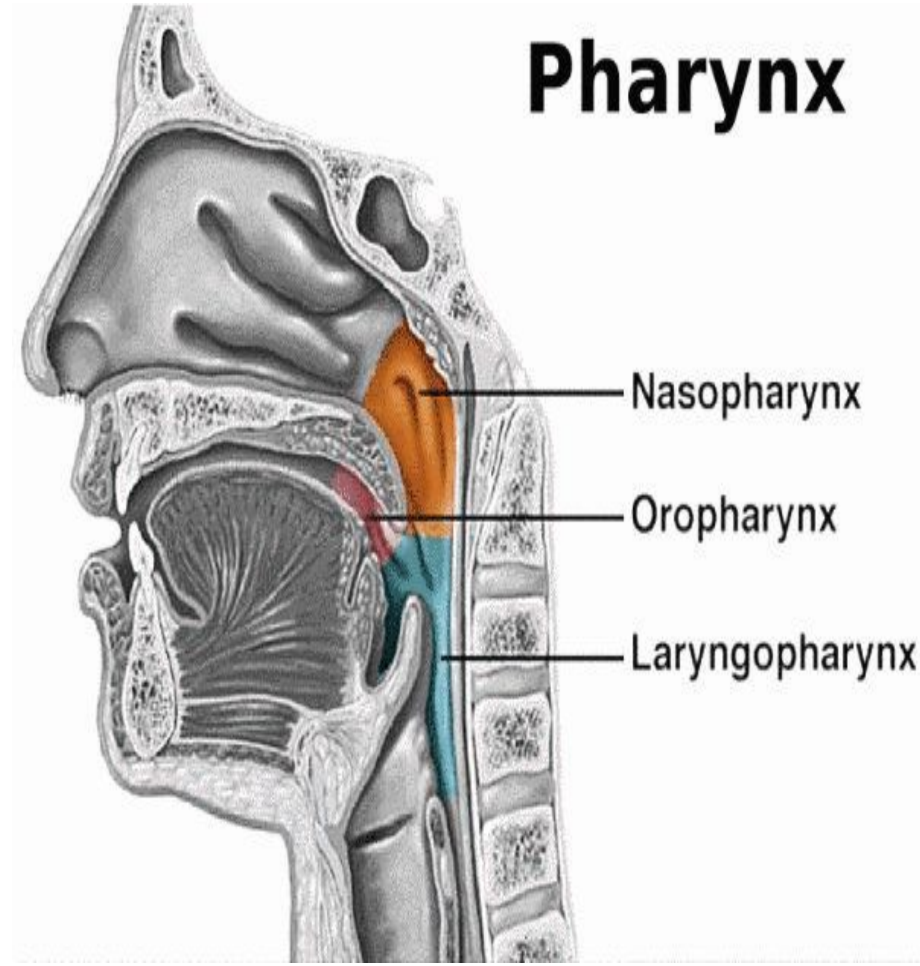
Introduction:

- It is a wide muscular tube located behind nose, mouth and larynx.
- About 12 cm in length.
- Its upper part transmits only air and the lower part only food but middle part transmits both air and food.



Boundaries:

- **Superiorly:** base of skull including body of sphenoid and basilar part of occipital bone.
- **Inferiorly:** continuous with the esophagus at the level of sixth cervical vertebra.
- **Posteriorly:** prevertebral fascia
- **Anteriorly:** communicates with the nasal cavity, oral cavity and larynx
- **On each side:** attached to medial pterygoid plate, pterygomandibular raphe, mandible, hyoid bone and thyroid and cricoid cartilage.



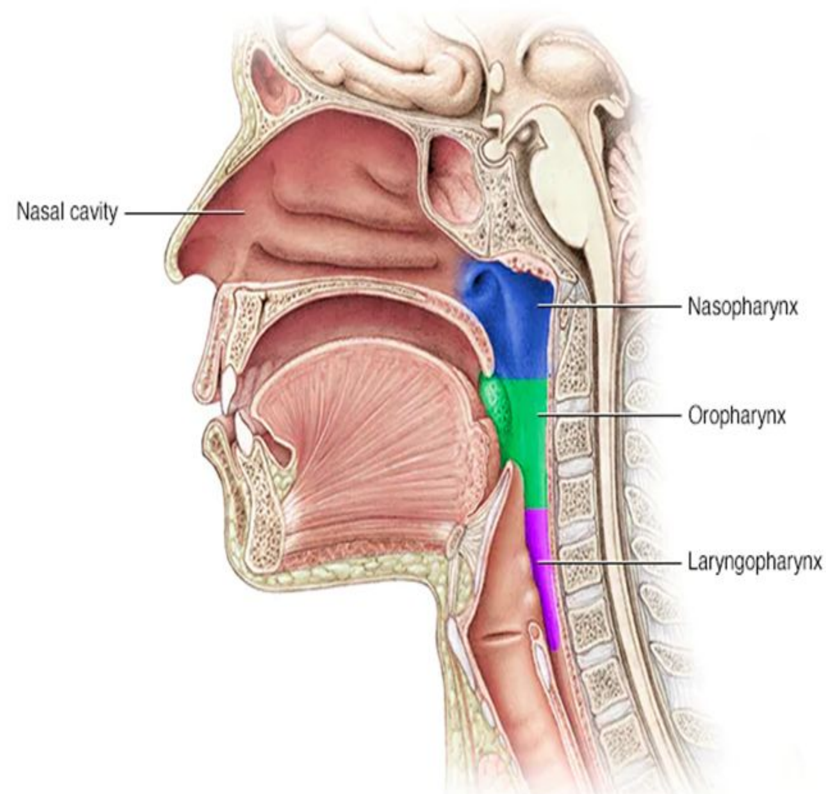
Parts of Pharynx:

- The pharynx is divided into:

a) Nasal part (nasopharynx)

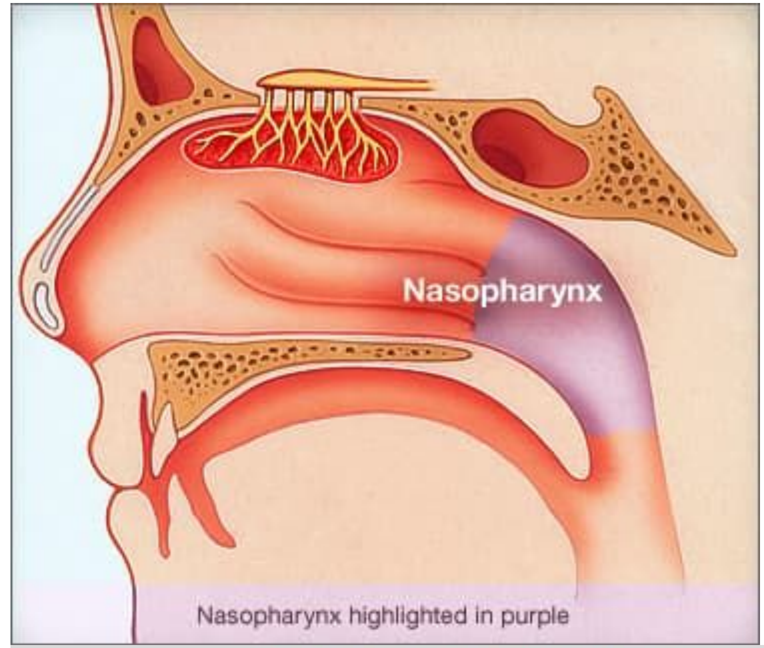
b) Oral part (oropharynx)

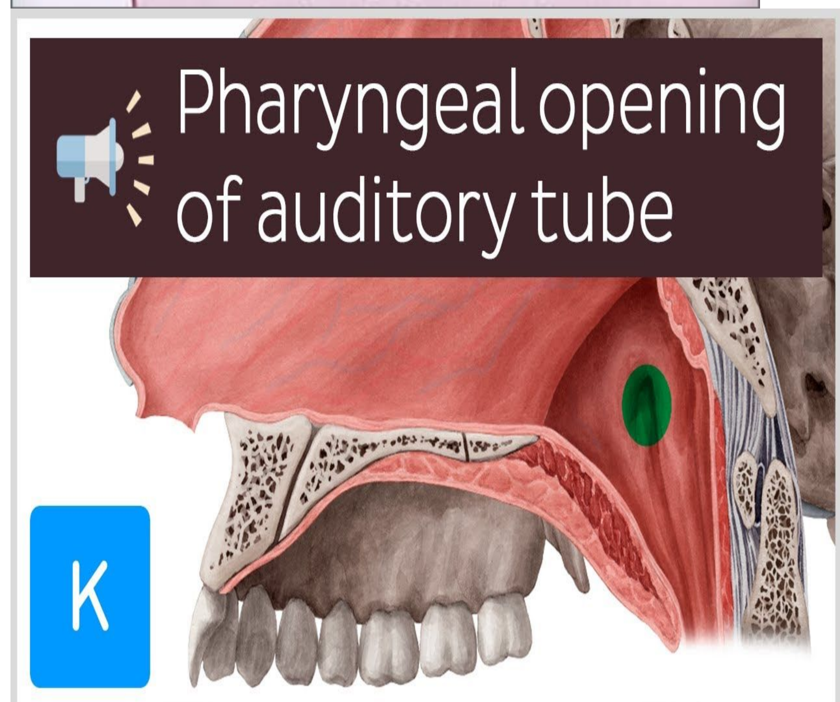
c) Larygeal part (laryngopharynx)



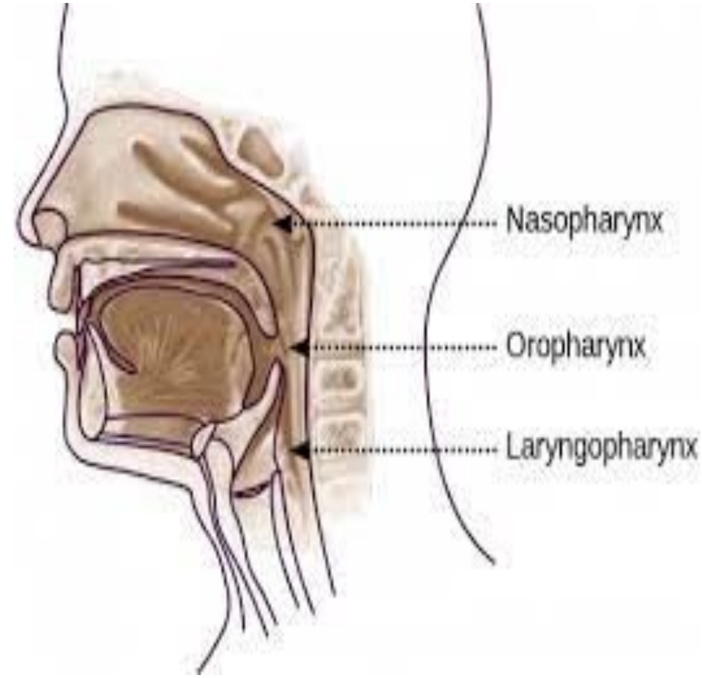
Nasopharynx:

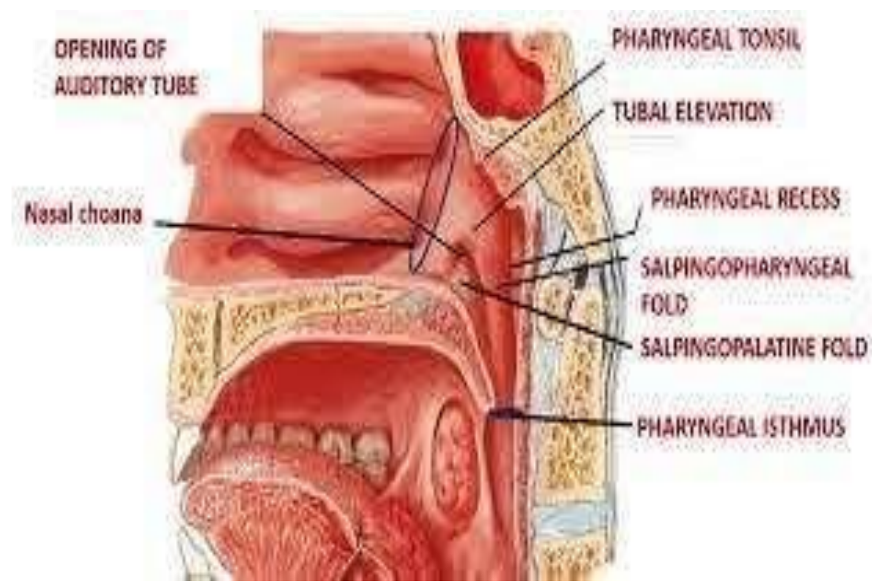
- Upper part of pharynx, situated behind nose and above lower border of soft palate.
- It is lined by pseudostratified ciliated columnar epithelium.
- Supplied by pharyngeal branch of pteryopalatine ganglion suspended by maxillary nerve.
- **Features of lateral wall:** it is formed by pharyngobasilar fascia and posterior median ligament.
- Laterally it communicates with the auditory tube and the tubal opening is surrounded by tubal elevation or torus tubaris.





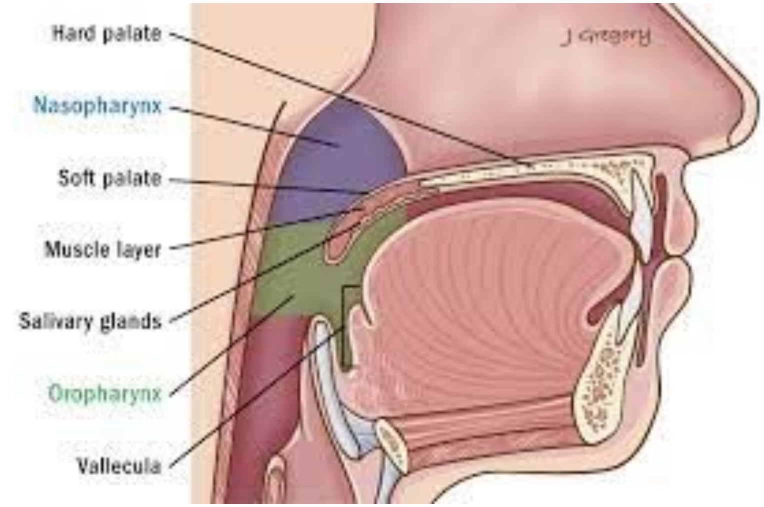
- Salpingopharyngeal fold is a vertical fold of mucous membrane containing salpingopharyngeus muscle.
- Salpingopalatine fold lies just below tubal elevation containing levator veli palatini muscle.
- **Roof/posterior wall:** present in the form of a slope against body of sphenoid, basiocciput and anterior arch of atlas.
- Pharyngeal tonsil lies under the mucous membrane opposite basiocciput.
- Hypertrophy of pharyngeal tonsil is termed as adenoids.

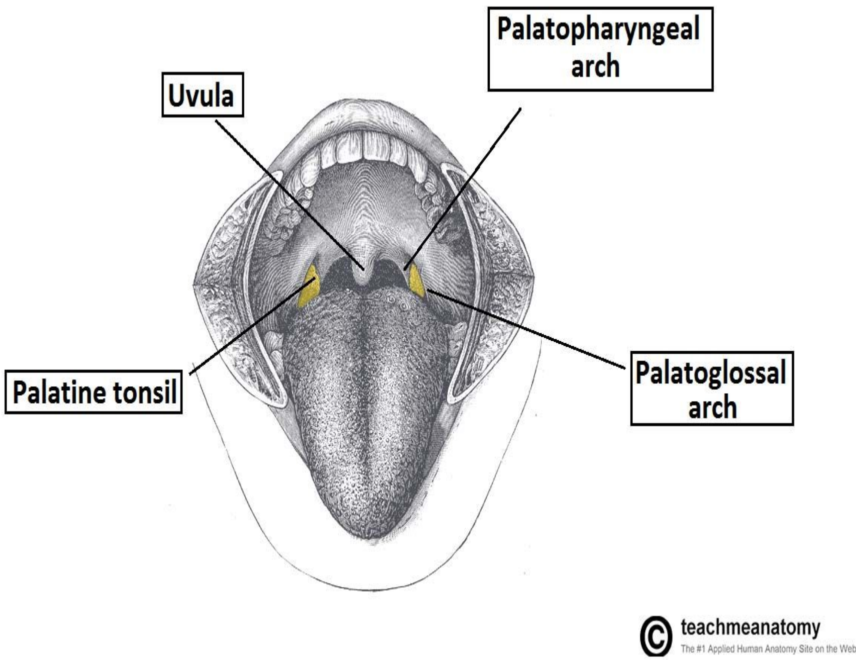




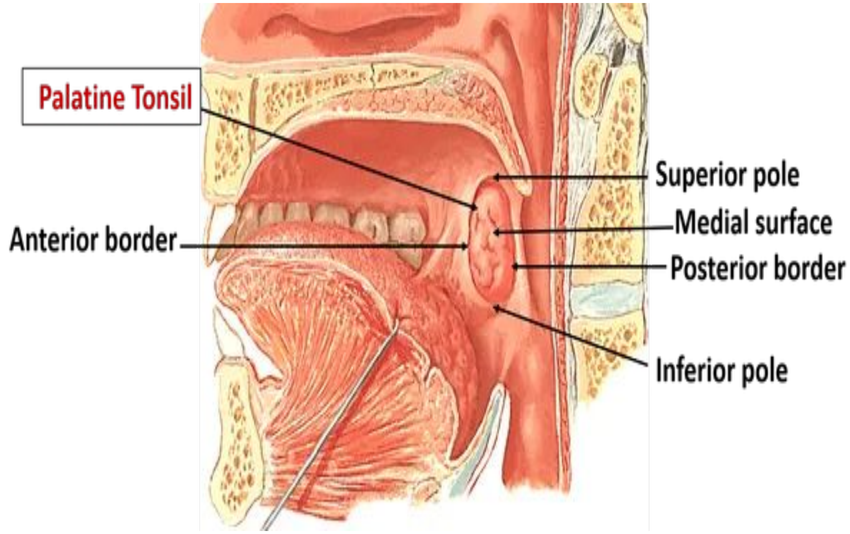
Oropharynx:

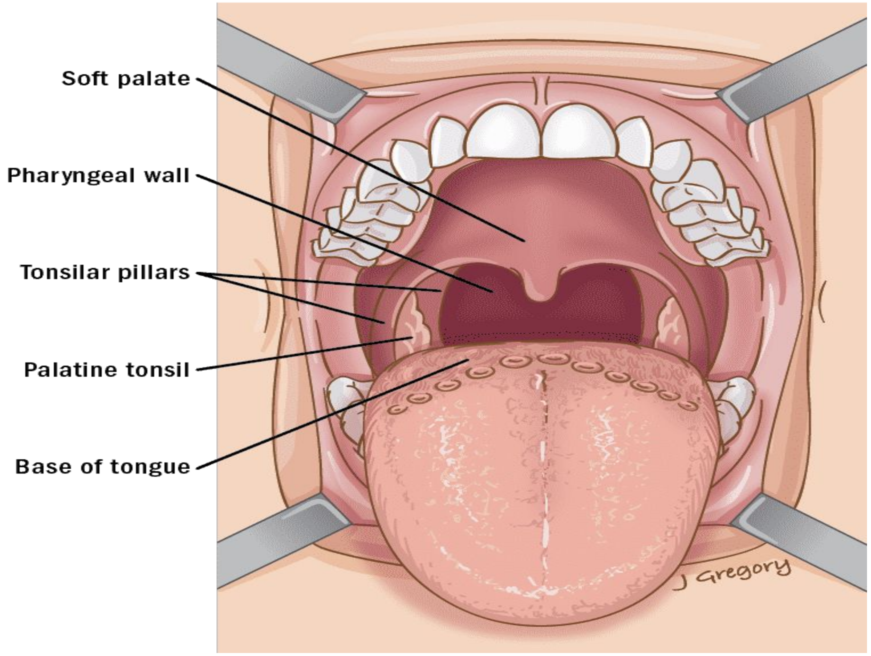
- It is the middle part situated behind oral cavity.
- Superiorly the oropharynx communicates with the nasopharynx at the oropharyngeal isthmus.
- inferiorly it communicates with the laryngopharynx at the upper border of epiglottis.
- Its lateral wall contains tonillar fossa/sinus occupied by palatine tonsil.





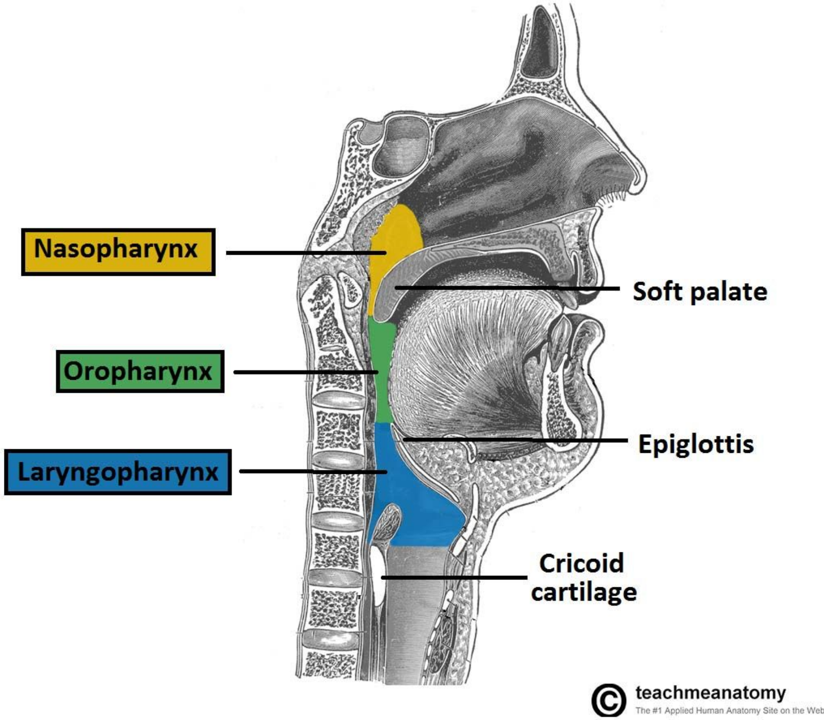
- **The palatine tonsil:** is almond-shaped.
- The tonsillar fossa is bounded anteriorly by palatoglossal arch and posteriorly by palatopharyngeal arch.
- The palatine tonsil has a medial surface covered by stratified squamous non-keratinized epithelium, &
- A lateral surface covered by capsule arising from pharyngobasilar fascia.
- The medial surface has 15-20 crypts, the largest one is called intratosillar cleft which is a site of peritonsillar abcess (quinsy).
- The palatine tonsil is supplied by tosillar branch of facial artery.
- Nerve supply by glossopharyngeal and lesser palatine nerves.

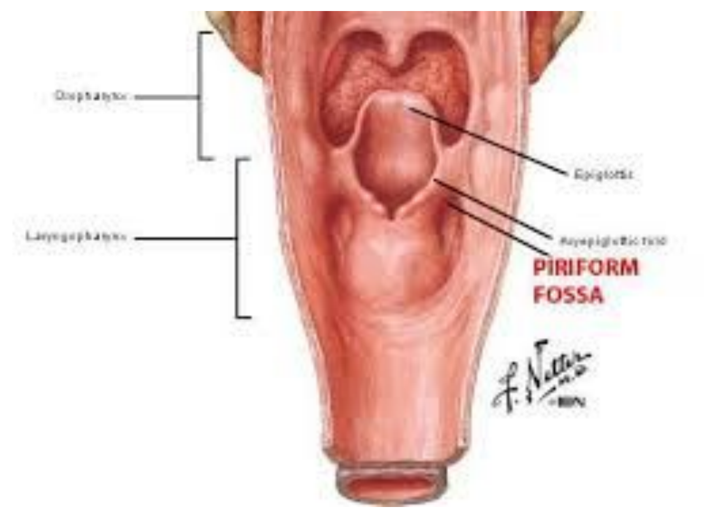




Laryngopharynx:

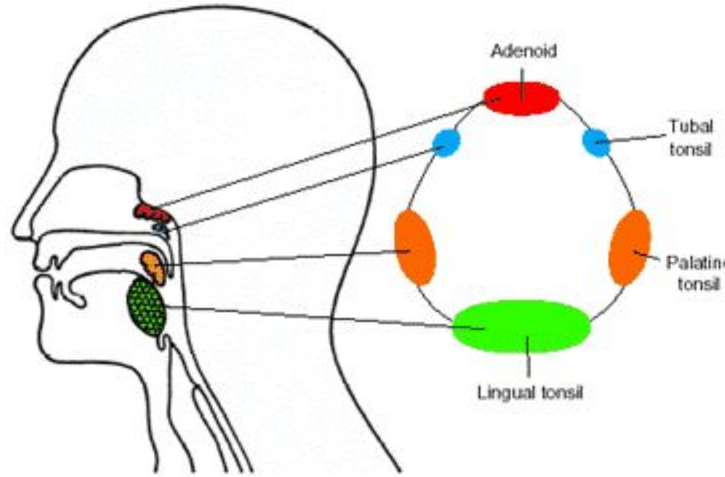
- Lower part of pharynx situated behind larynx.
- **Extent:** from upper border of epiglottis to lower border of cricoid cartilage.
- Its lateral wall has a depression called piriform fossa.

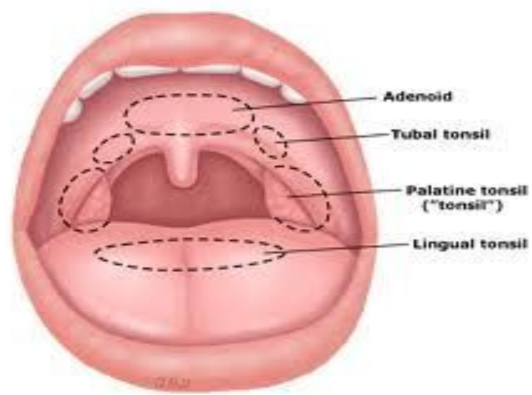




Waldeyer’s lymphatic ring:

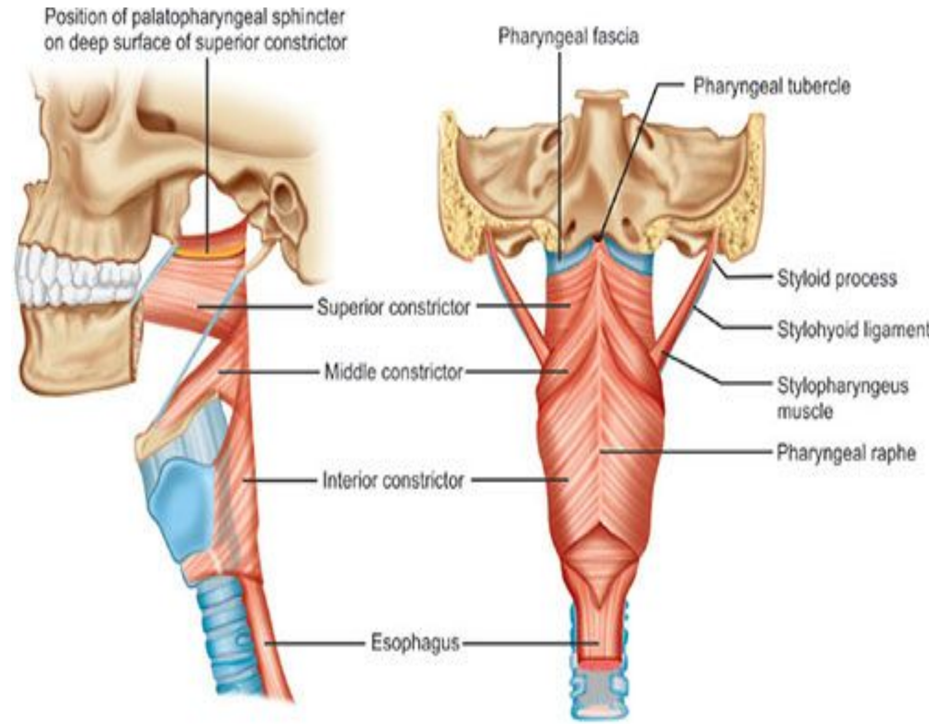
- At the level of orpharyngeal isthmus, there are some aggregations of lymphoid tissue that form Waldeyer’s lymphatic ring.

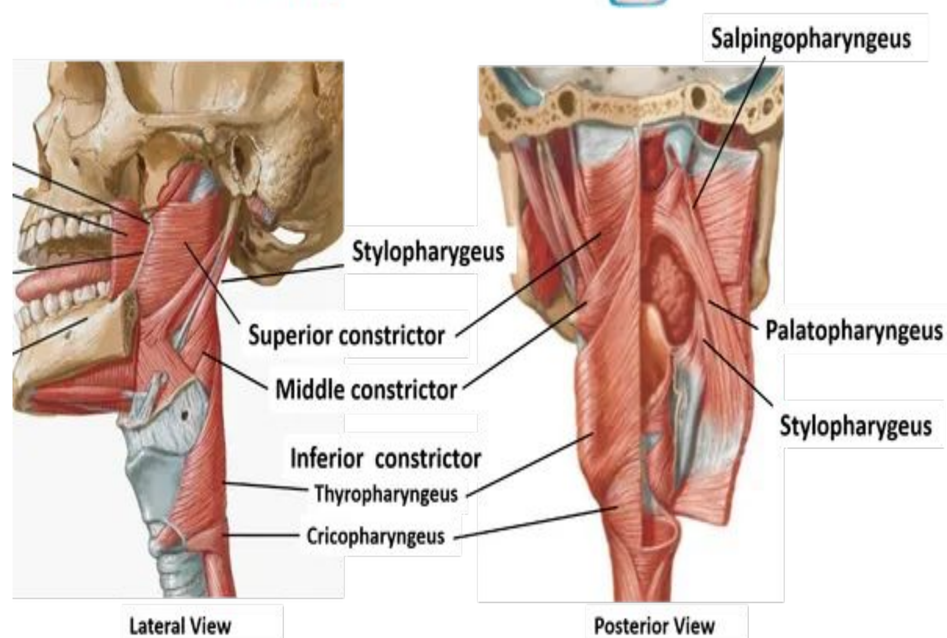


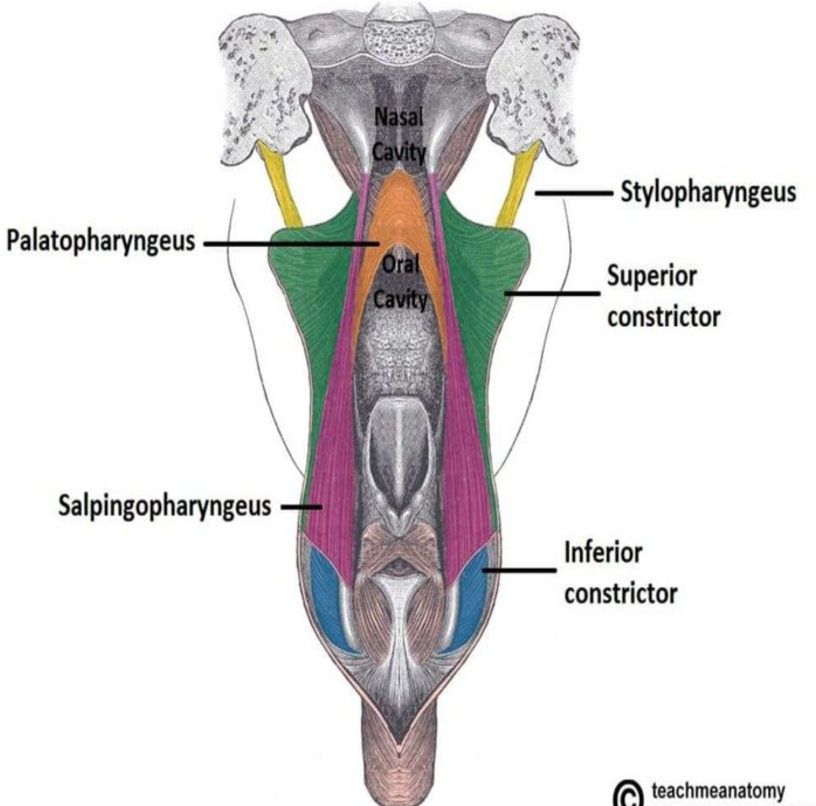


Muscles of Pharynx:

- Muscular coat of pharynx consists of:
- An outer circular layer made up of three constrictors i.e. **superior, middle** and **inferior,** &
- An inner longitudinal layer comprising of **stylopharyngeus, salpingopharyngeus** and **palatopharyngeus** muscles.






Nasal Cavity
Stylopharyngeus
Palatopharyngeus
Oral Cavity
Superior constrictor
Salpingopharyngeus
Inferior constrictor
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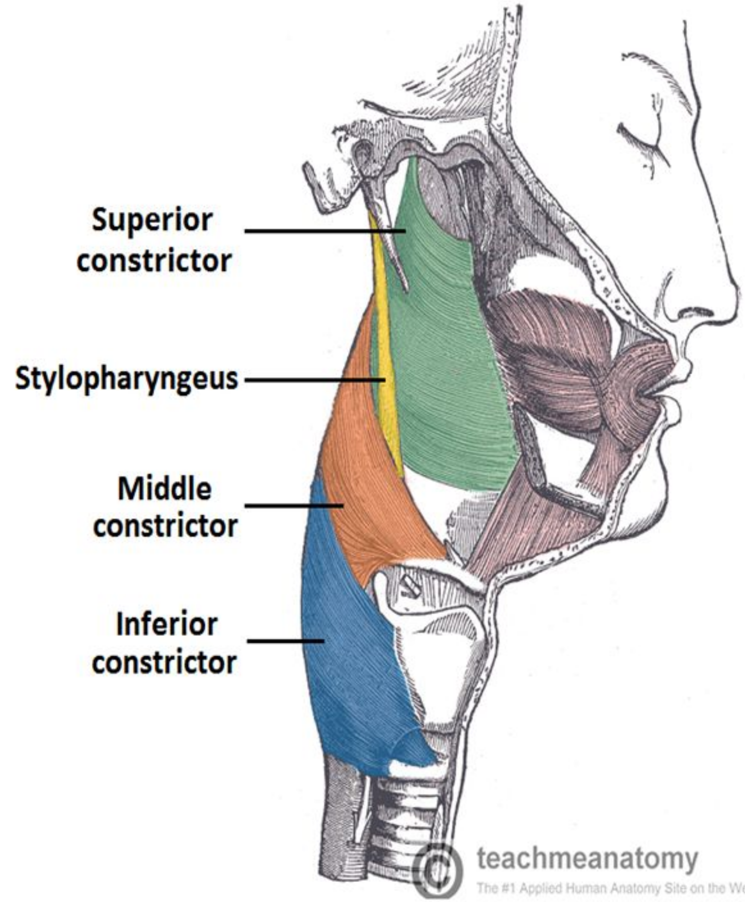
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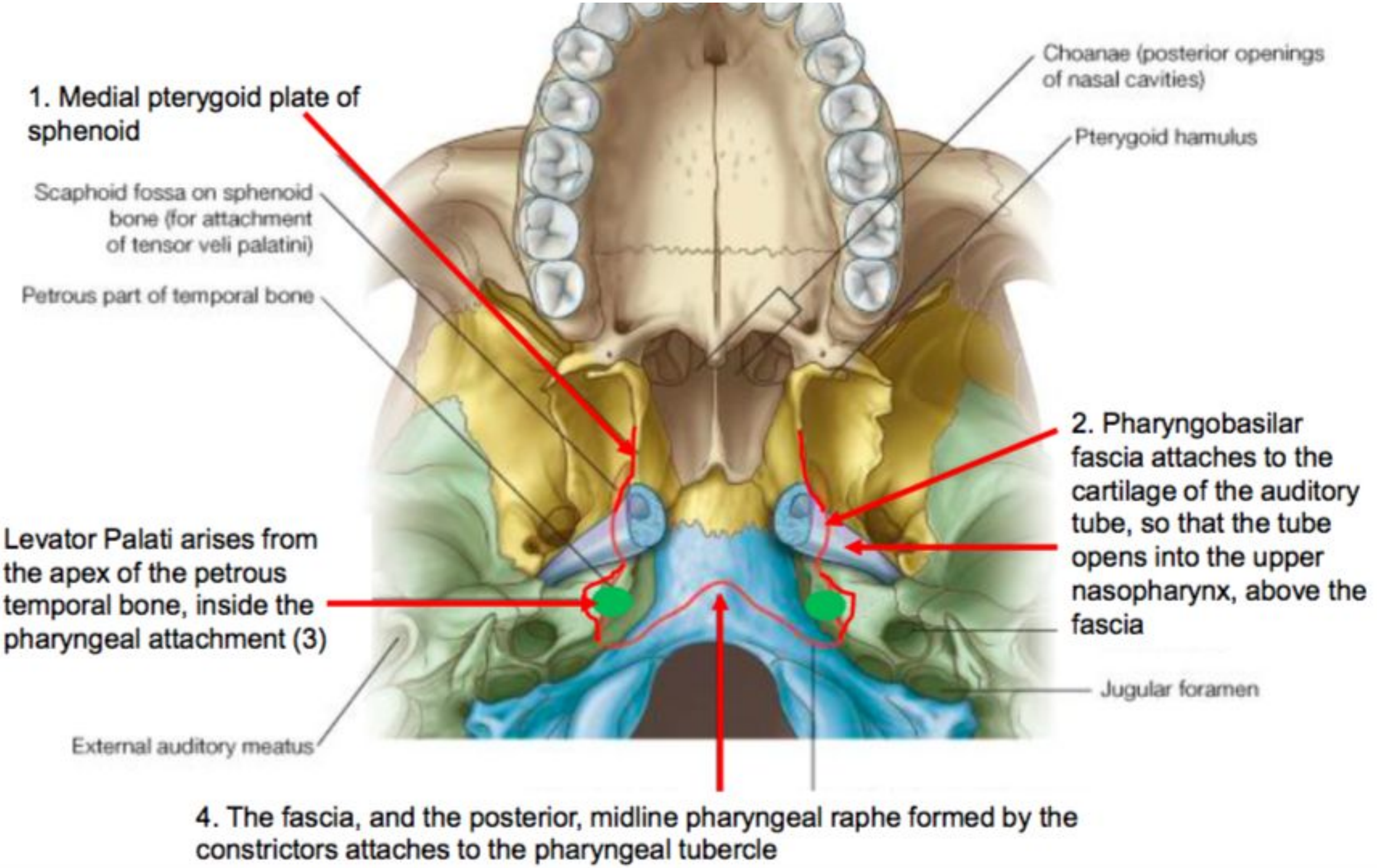

1. Medial pterygoid plate of sphenoid
Scaphoid fossa on sphenoid bone (for attachment of tensor veli palatini)
Petrous part of temporal bone
Levator Palati arises from the apex of the petrous temporal bone, inside the pharyngeal attachment (3)
External auditory meatus
Choanae (posterior openings of nasal cavities)
Pterygoid hamulus
2. Pharyngobasilar fascia attaches to the cartilage of the auditory tube, so that the tube opens into the upper nasopharynx, above the fascia
Jugular foramen
4. The fascia, and the posterior, midline pharyngeal raphe formed by the constrictors attaches to the pharyngeal tubercle

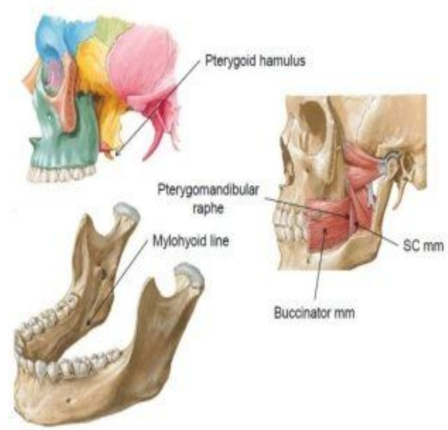
Constrictor muscles of pharynx:

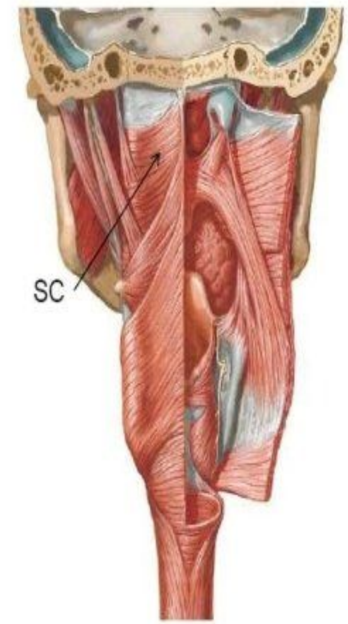
- All the constrictor muscles are inserted into median raphe on posterior wall of pharynx.
- All the muscles of pharynx are supplied by pharyngeal plexus except for stylopharyngeus muscle which is supplied by glossopharyngeal nerve.

Superior constrictor muscle

Origin (from above to downwards)

- Pterygoid hamulus
- Pterygomandibular raphe
- Medial surface of mandible at posterior end of mylohyoid line (near the attachment of pterygomandibular raphe)
- Side of the posterior part of the tongue

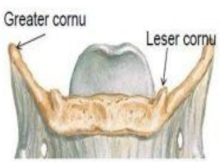


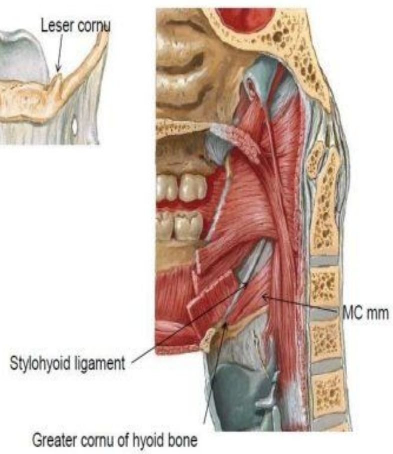


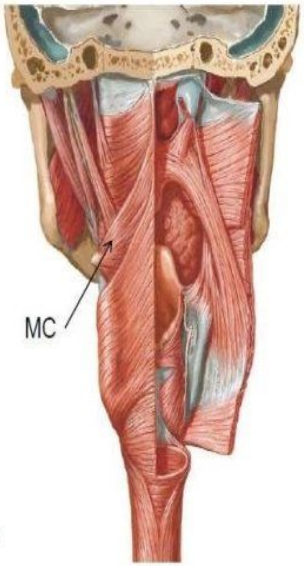
Middle constrictor muscle

Origin

- Lower part of stylohyoid ligament
- Lesser cornu of hyoid bone
- Upper border of greater cornu of hyoid bone

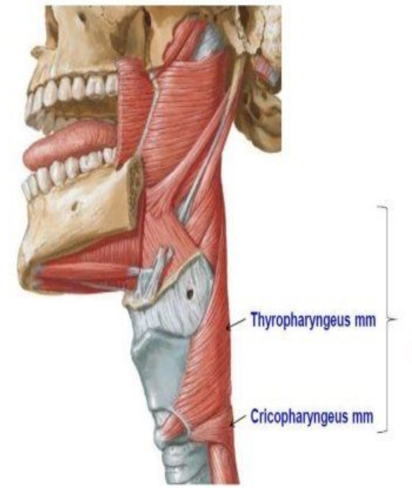






Inferior constrictor muscle

1. Thyropharnygeus
   - Arise from **thyroid cartilage** –oblique line and inferior tubercle of thyroid cartilage
   - A tendinous band that crossed the cricothyroid muscle & is attached *above to the inferior tubercle of thyroid cartilage*
1. Cricopharyngeus
   - Cricoid cartilage behind the origin of cricothyroid muscle

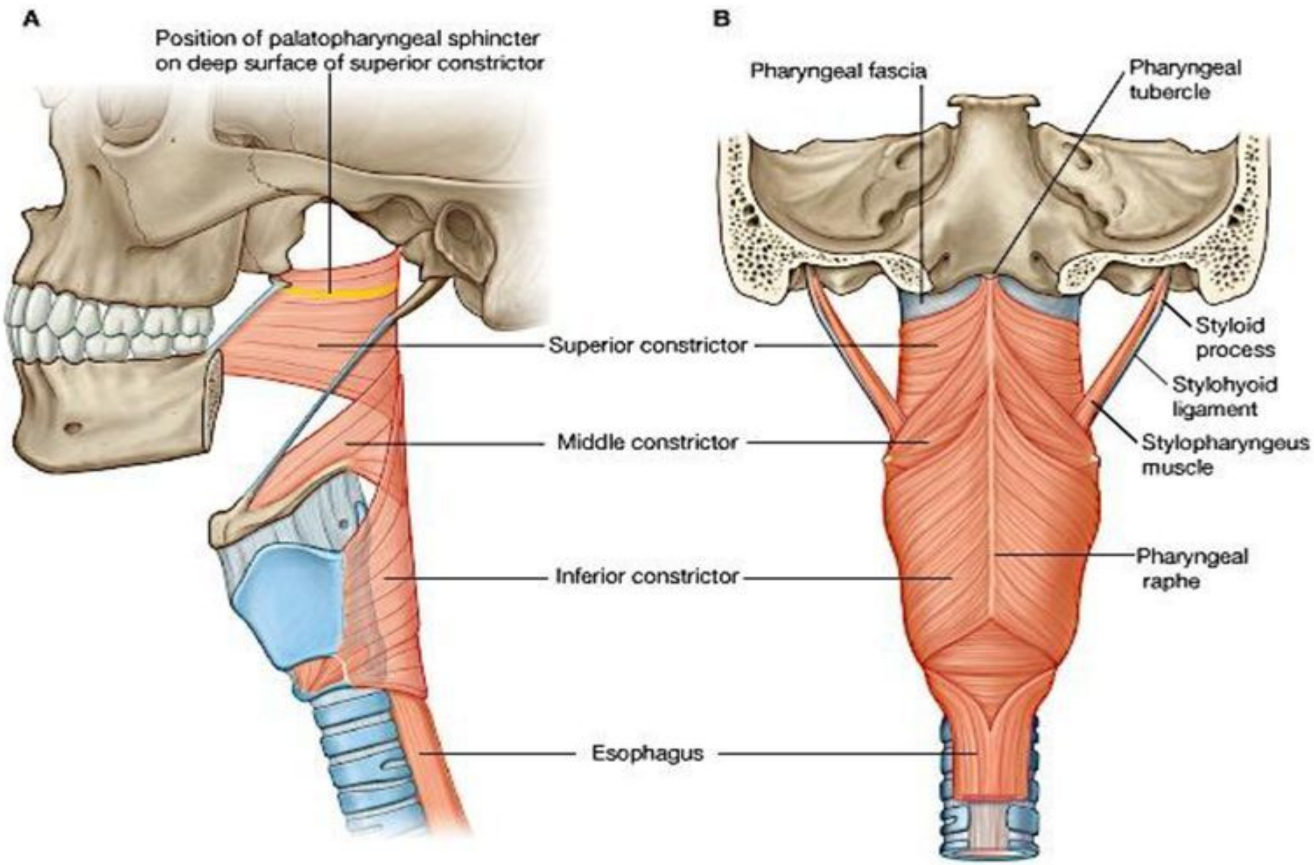


Longitudinal Muscles of The Pharynx

Muscle	Origin	Insertion	Nerve supply
Stylopharyngeus	Medial surface of the base of styloid process	Posterior border of the lamina of thyroid cartilage	Glossopharyngeal (IX) nerve
Palatopharyngeus	By two fasciculi (anterior and posterior) from the upper surface of the palatine aponeurosis	Posterior border of the lamina of thyroid cartilage	Cranial root of 11th cranial nerve by pharyngeal plexus
Salpingopharyngeus	Lower part of the cartilage of the auditory tube	Posterior border of the lamina of thyroid cartilage	Cranial root of 11th cranial nerve by pharyngeal plexus

- **Separated into 2 layers, pharyngeal muscles sandwiched between them**
- **Thick pharyngobasilar fascia inside**
- **Thin buccopharyngeal fascia outside**
- **Reinforces the pharyngeal wall**

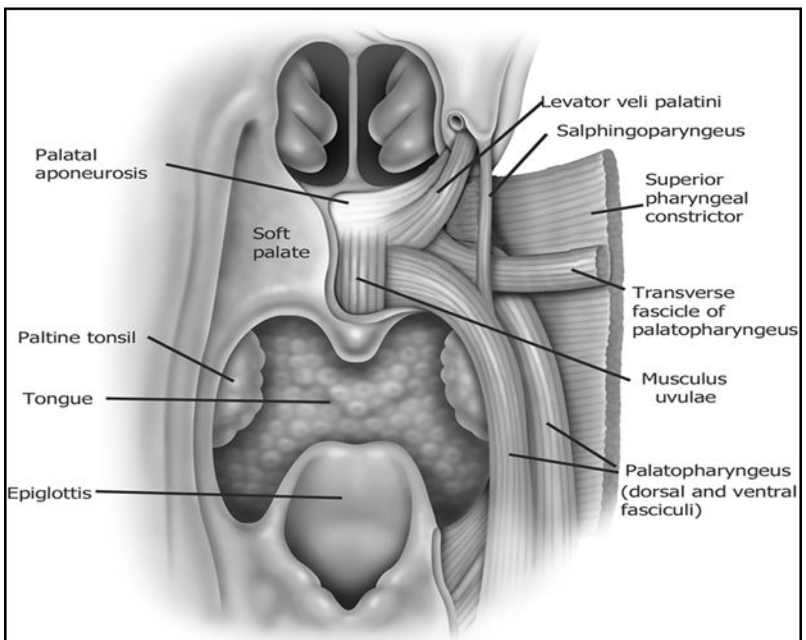
Pharyngeal Fascia



Function of muscles of pharynx:

Longitudinal muscles:

- They act to shorten and widen the pharynx, and elevate the larynx during swallowing.



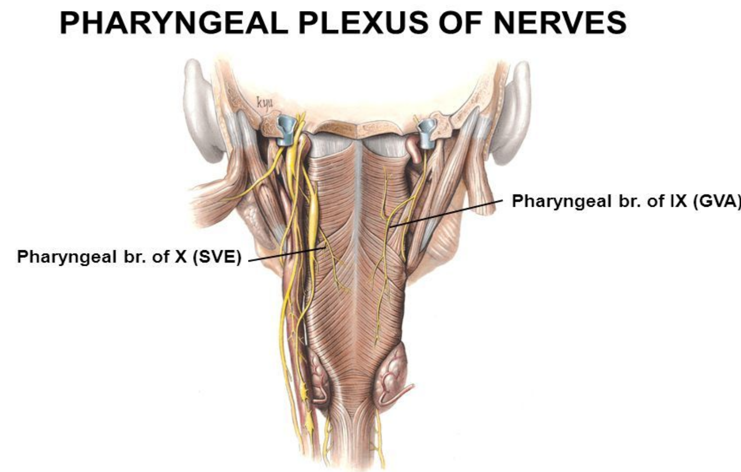
Circular/constrictor muscles:

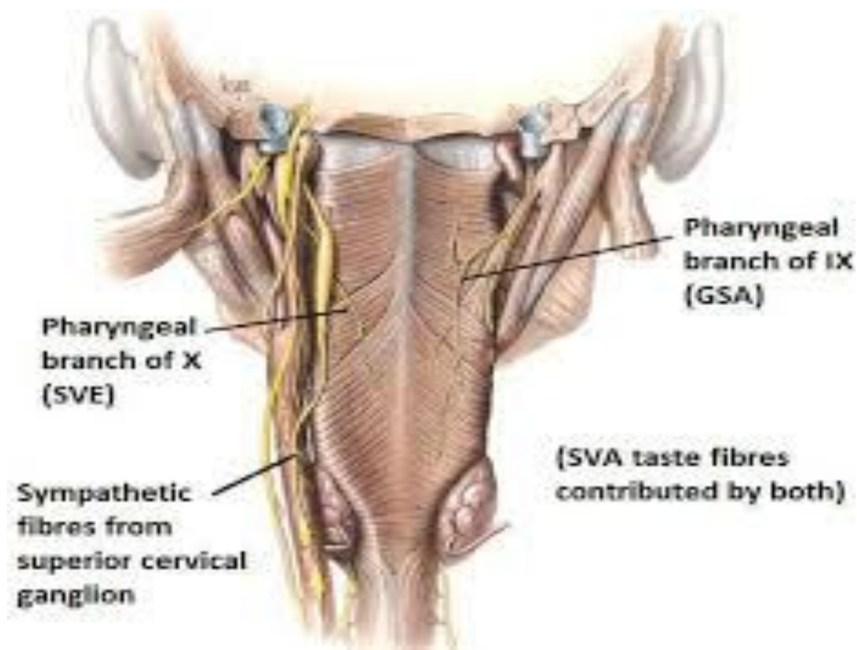
- The circular muscles contract sequentially from superior to inferior to constrict the pharyngeal lumen. and thus propel the bolus of food inferiorly into the oesophagus.

Pharyngeal plexus:

- The pharyngeal plexus of nerves is formed by:

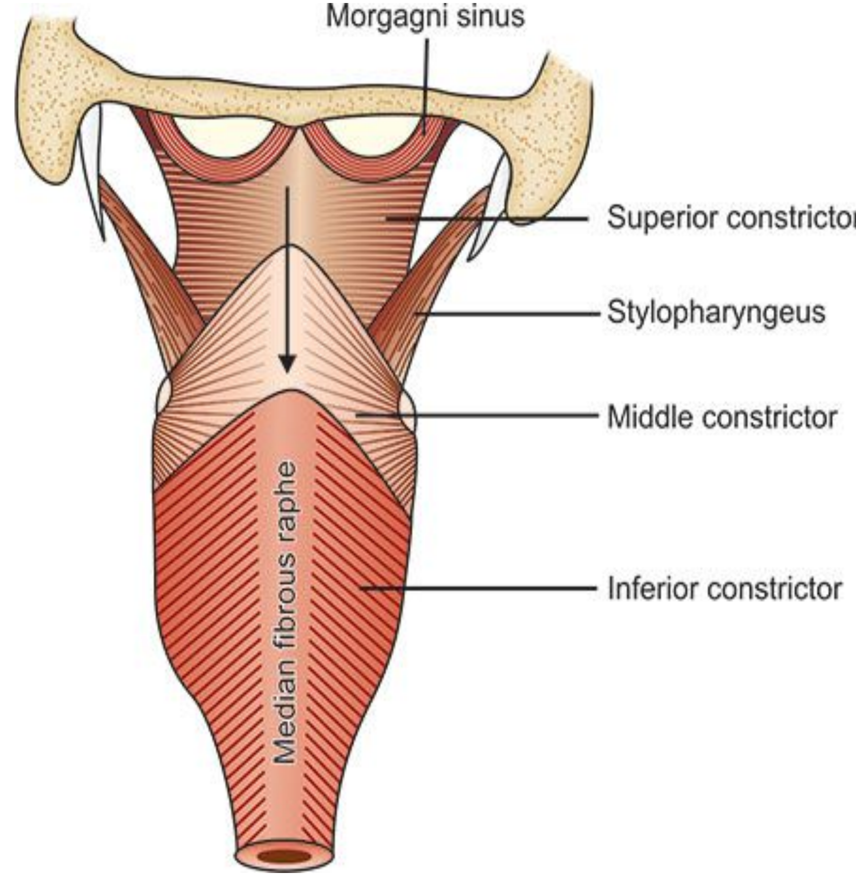
i. Pharyngeal branch of vagus carrying fibres of cranial accessory nerve
ii. Pharyngeal branches of glossopharyngeal nerve
iii. Pharyngeal branches of superior cervical sympathetic ganlion





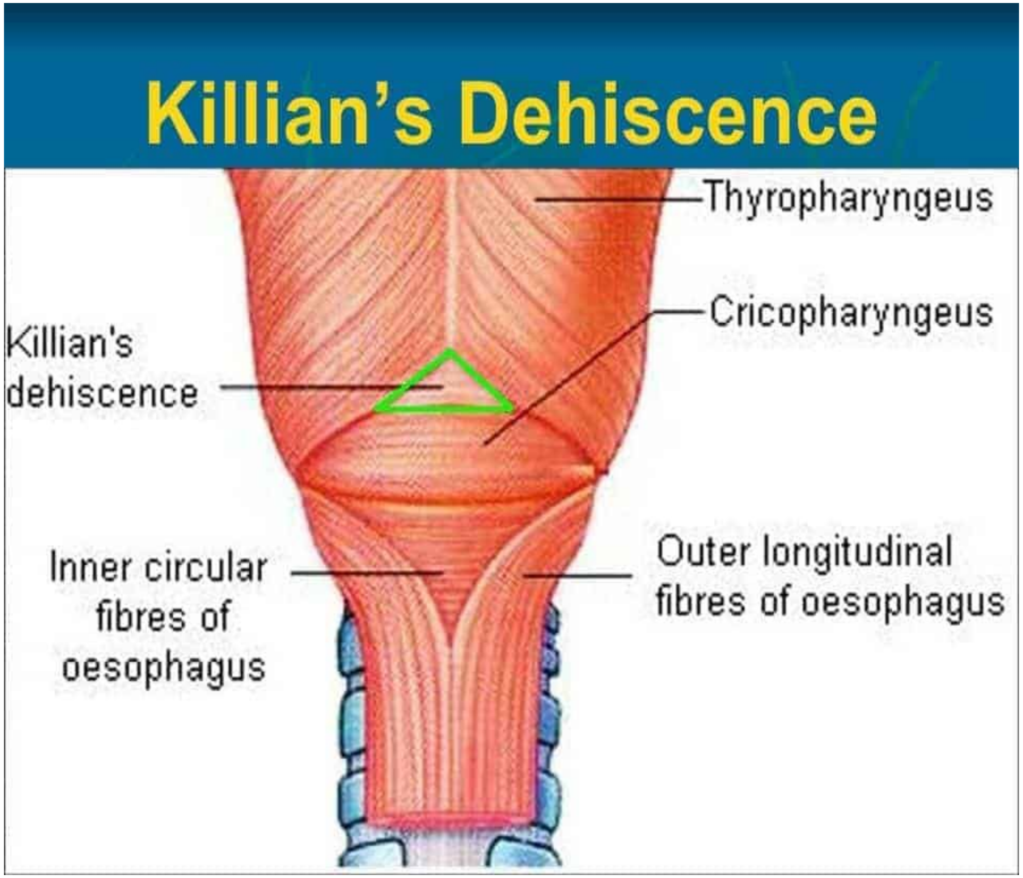
Gaps between pharyngeal muscles:

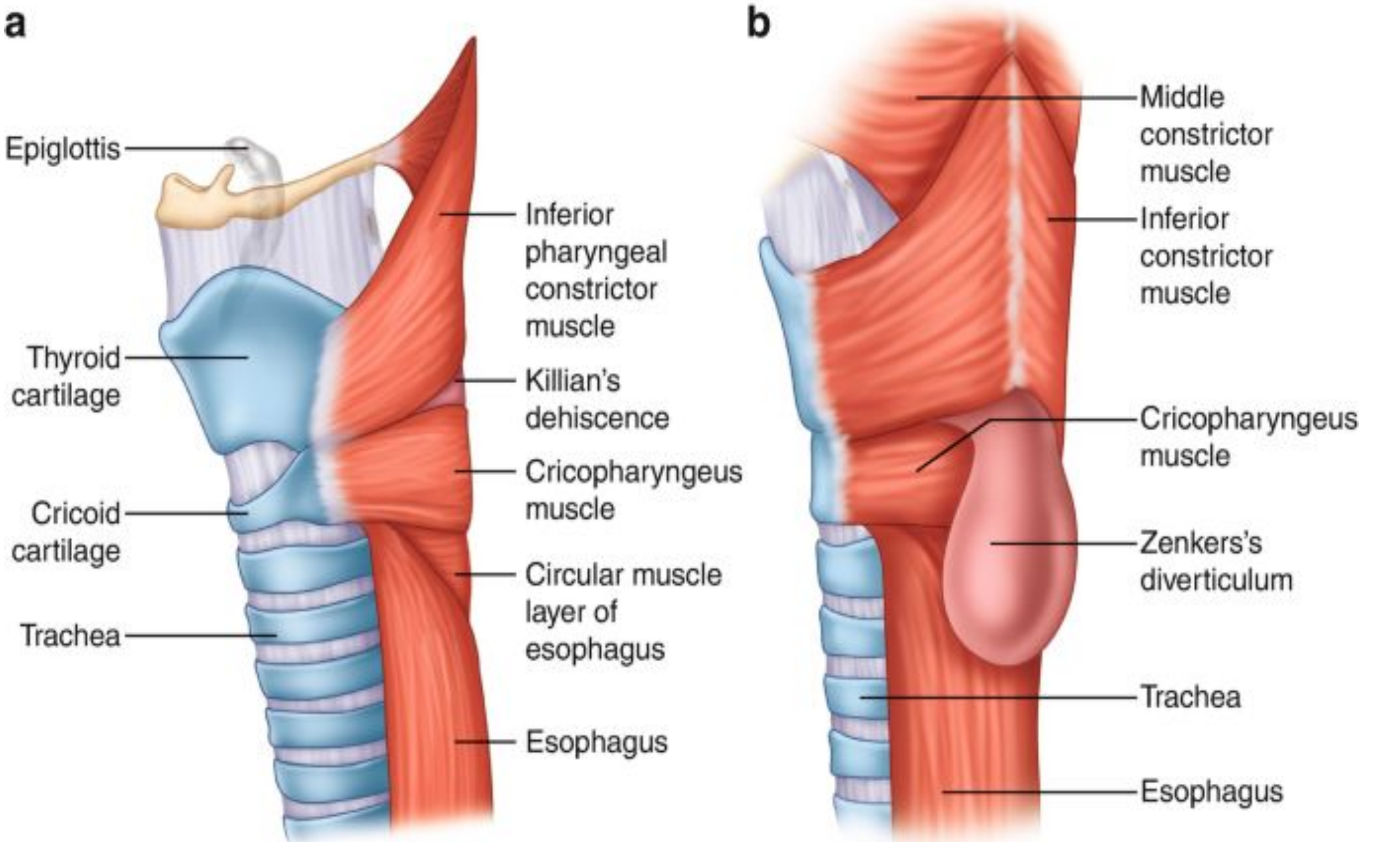
1) **Sinus of Morgagni:** is the gap between upper border of superior constrictor and base of the skull.
   - It is closed by pharyngobasilar fascia.
   - Structures passing through this gap are: auditory tube, levator veli palatini muscle and ascending palatine artery.



2) **Killian's Dehiscence:** is a lower part of thyropharyngeus muscle which is not internally supported by upper or middle constrictors.

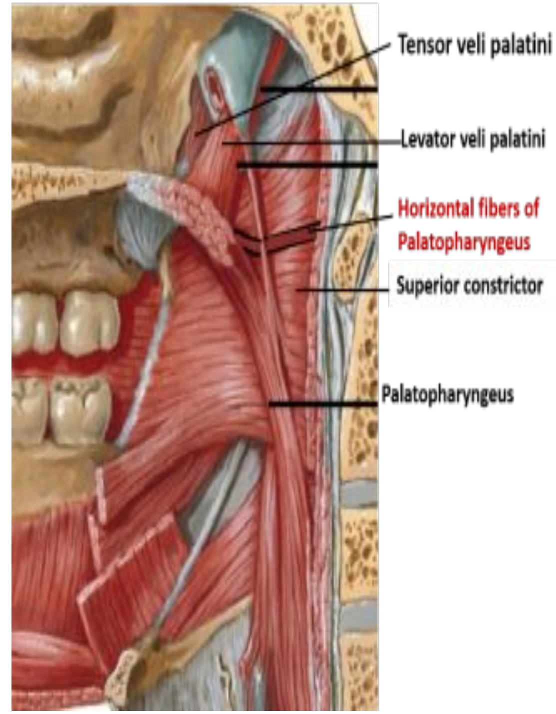
- This weak part lies below the level of vocal cords and is a common site of pharyngeal diverticula.

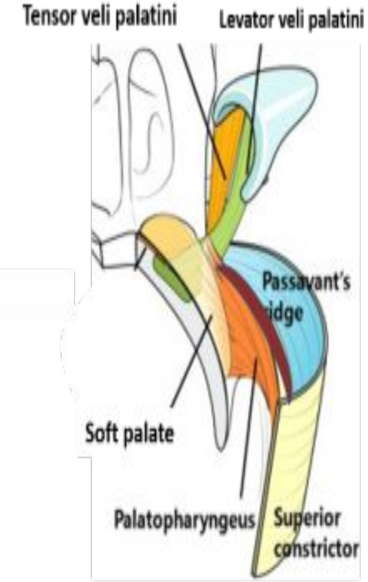


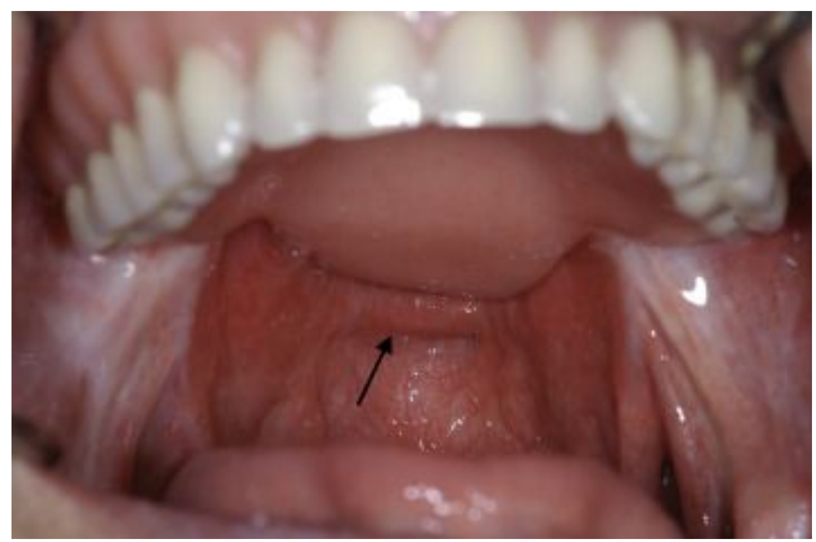

a
Epiglottis
Thyroid cartilage
Cricoid cartilage
Trachea
Inferior pharyngeal constrictor muscle
Killian's dehiscence
Cricopharyngeus muscle
Circular muscle layer of esophagus
Esophagus
b
Middle constrictor muscle
Inferior constrictor muscle
Cricopharyngeus muscle
Zenkers's diverticulum
Trachea
Esophagus

Passavant’s Ridge:

- Some fibres of palatopharyngeus muscle, arranged circularly deep to mucous membrane form a sphincter internal to superior constrictor muscle.
- These fibres form passavant’s muscle.
- This muscle contracts to raise a ridge called passavant’s ridge on the posterior wall of pharynx.
- When soft palate is elevated, it comes in contact with the ridge, thus closing the pharyngeal isthmus between nasopharynx and oropharynx.





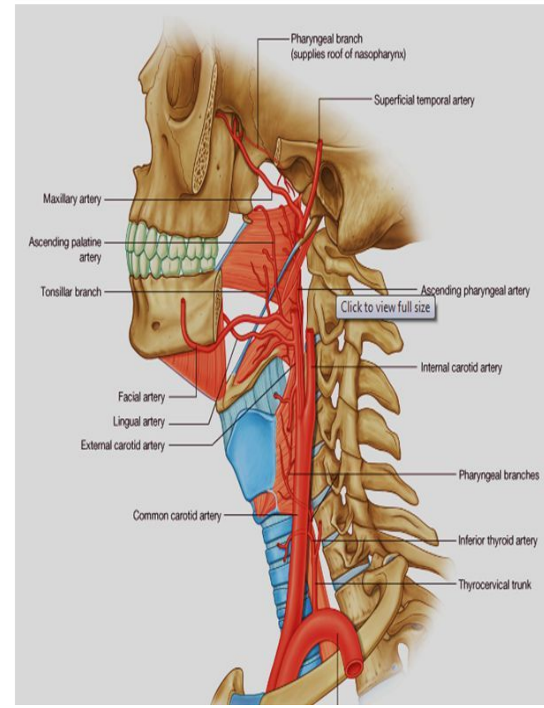


Blood supply and nerve supply of pharynx:

- Pharynx is supplied by branches of external carotid, facial, lingual and maxillary arteries.
- General sensory supply mostly via glossopharyngeal and partly by vagus nerve.
- Parasympathetic/secretomotor supply via greater petrosal branch of facial nerve lesser palatine branch of pterygopalatine ganglion.

Blood supply of the Pharynx

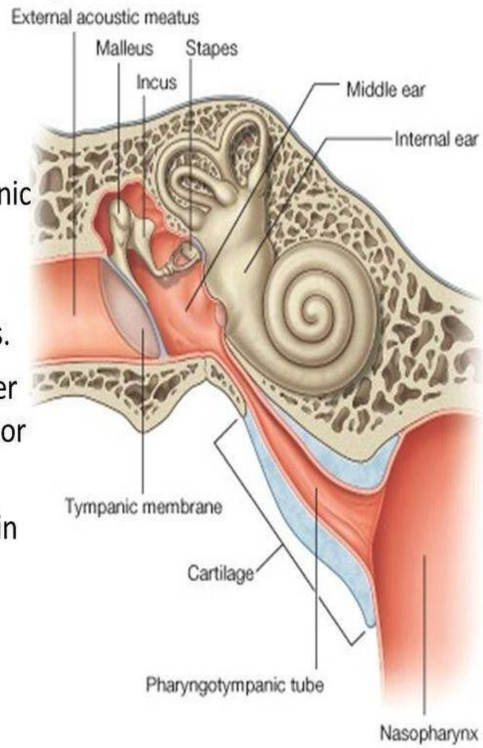
- **Arteries:**
- **Ascending pharyngeal artery , external carotid artery medial group**
- **Ascending palatine artery , from facial artery of external carotid artery**
- **Tonsillar artery ,**
- **Maxillary artery**
- **Lingual artery**



The Eustachian/Auditory Tube:

Auditory Tube

- Also called **Eustachian Tube or Pharyngotympanic tube**
- It connects anterior wall of tympanic cavity to nasal pharynx.
- Its posterior third is bony and its anterior two thirds is cartilaginous.
- As the tube descends it passes over upper border of superior constrictor muscle of pharynx.
- It serves to equalize air pressures in the tympanic cavity and the nasal pharynx.



- **Blood supply:** it is supplied by the ascending pharyngeal artery and middle meningeal artery and the artery of pterygoid canal.
- **Nerve supply:** cartilaginous part by nervus spinosus from mandibular nerve and bony part by tympanic plexus formed by glossopharyngeal nerve.